

=== CONFIGURACION GNB (1779879009) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an TDD cell with 20 MHz bandwidth
Optimized for low latency and gaming scenarios for a LAN Party.

```
cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF (Open5GS container IP).
    port: 38412
    bind_addr: 10.53.1.4     # A local IP that the gNB binds to for traffic from the AMF (gNB container IP).
    supported_tracking_areas:
      - tac: 7
    plmn_list:
      - plmn: "00101"        # PLMN for University of Granada LAN Party (MCC 001, MNC 01).
    tai_slice_support_list:
      - sst: 1               # SST 1 for eMBB, suitable for low-latency gaming.
    inactivity_timer: 7200   # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:
  device_driver: zmq         # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=23.04e6 # ZMQ ports for gNB.
  srate: 23.04              # RF sample rate adjusted for 20 MHz bandwidth.
  tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75               # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 620000          # ARFCN of the downlink carrier (center frequency) for Band 78.
  band: 78                  # The NR band (Band 78, n78).
  channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 15            # Subcarrier spacing in kHz used for data and SSB.
  plmn: "00101"            # PLMN broadcasted by the gNB.
  tac: 7                    # Tracking area code (needs to match the core configuration).
  pucch:
    sr_period_ms: 20        # SR period 20ms for faster UL scheduling requests, beneficial for gaming.
    nof_cell_sr_res: 50
    nof_cell_csi_res: 50
  csi:
    csi_rs_period: 40       # CSI period 40ms.
  ul_common:
    max_ul_grants_per_slot: 64 # Max UL grants per slot for efficient UL resource allocation.
    max_pucch_per_slot: 63    # Max PUCCHs per slot.
  pusch:
    max_pusch_per_slot: 1    # Max PUSCHs per slot.
  tdd_ul_dl_cfg:
    nof_dl_symbols: 7
    nof_dl_slots: 7
    nof_ul_slots: 2          # TDD configuration optimized for UL, good for gaming.
  pdccch:
    common:
      ss0_index: 0           # Set search space zero index to match srsUE capabilities
      coreset0_index: 12     # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common       # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1    # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64         # Sets PDSCH MCS to 64 QAM for higher throughput.
  pushe:
    mcs_table: qam64         # Sets PUSCH MCS to 64 QAM for higher throughput.

log:
  filename: /tmp/gnb.log     # Path of the log file.
  all_level: info           # Logging level applied to all layers.
  hex_max_size: 0
```

pcap:

```
mac_enable: false           # Set to true to enable MAC-layer PCAPs.
mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
ngap_enable: false          # Set to true to enable NGAP PCAPs.
ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
e2ap_enable: true           # Set to true to enable E2AP PCAPs.
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.
```

metrics:

layers:

```
enable_rlc: true           # Enable RLC metrics
enable_sched: true         # Enable DU scheduler metrics
```

periodicity:

```
du_report_period: 1000     # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779879009) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 23.04e6          # RF sample rate matching gNB for 20 MHz bandwidth.
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=23.04e6 # ZMQ ports inverse to gNB.

[rat.eutra]
dl_eearfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78                # NR Band 78, matching gNB.
nof_carriers = 1
dl_nr_arfcn = 620000      # NR ARFCN matching gNB.
max_nof_prb = 106        # Max PRBs for 20 MHz bandwidth, 15 kHz SCS.
nof_prb = 106            # Number of PRBs for 20 MHz bandwidth, 15 kHz SCS.

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc  = 63BFA50EE6523365FF14C1F45F88737D
k    = 00112233445566778899aabbccddeeff
imsi = 001010123456780    # IMSI matching PLMN (MCC 001, MNC 01) for UGR LAN Party.

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn              # Access Point Name.
apn_protocol = ipv4

[gw]
#netns = ue1
#ip_devname = tun_srsue
#ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER COMPOSE (1779879009) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    env_file:
      - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
    privileged: true
    ports:
      - "9898:9999/tcp"
      # Uncomment port to use the 5gc from outside the docker network
      # - "38412:38412/sctp"
      # - "2152:2152/udp"
    command: 5gc -c open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2} # Open5GS Core IP
```

gnb:

```
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  # privileged mode is required only for accessing usb devices
  privileged: true
  # Extra capabilities always required
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    # Access USB to use some SDRs
```

```

- /dev/bus/usb/:/dev/bus/usb/
# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
- gnb_config.yml
- gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.4} # gNB container IP, avoiding 10.53.1.3 for binds.
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

```

```

ue:
  container_name: srsran_ue
  image: srsran/ue
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  privileged: true
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    - /dev/bus/usb/:/dev/bus/usb/
    - ue-storage:/tmp
  configs:
    - ue_config.conf
  networks:
    ran:
      ipv4_address: ${UE_IP:-10.53.1.5} # UE container IP.
  depends_on:
    gnb:
      condition: service_healthy
  command: ue -c /ue_config.conf

```

```

configs:
  gnb_config.yml:
    file: ${GNB_CONFIG_PATH:-./configs/gnb.yml} # Path to your desired gNB config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2} # AMF IP matching Open5GS container.
          bind_addr: ${GNB_IP:-10.53.1.4} # gNB bind IP matching gNB container.
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true
  ue_config.conf:
    file: ${UE_CONFIG_PATH:-./configs/ue.conf} # Path to your desired UE config file

```

volumes:

gnb-storage:

ue-storage:

networks:

ran:

ipam:

driver: default

config:

- subnet: 10.53.1.0/24

metrics:

ipam:

driver: default

config:

- subnet: 172.19.1.0/24